

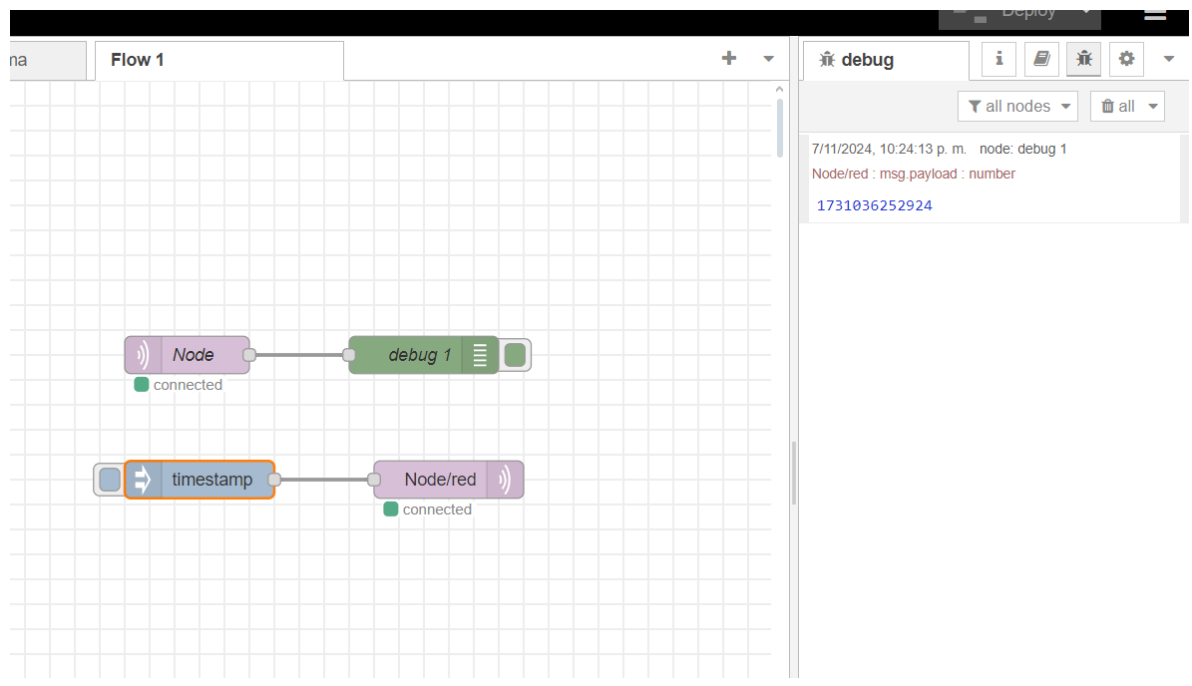
EVIDENCIAS PRACTICA NODE RED

Integrantes: Daniel Gustavo Rojas Hernández

Fecha: 07/11/2024

Actividad 1:

Pegar aquí los pantallazos de los resultados que evidencien la recepción de mensajes en el nodo **debug** en node red



Recuperar Flujoograma

Edit mqtt in node

Delete Cancel Done

Properties

Server: EMQX

Action: Subscribe to single topic

Topic: Node/red

QoS: 0

Flags: ☐ Do not receive messages published by this client ☒ Keep retain flag of original publish

Retained message handling: Send retained messages

Output: auto-detect (parsed JSON object, string or buffer)

Name: Node

debug

all nodes all

7/11/2024, 10:24:13 p. m. node: debug 1

Node/red : msg.payload : number

1731036252924

Flow

Edit mqtt out node

Delete Cancel Done

Properties

Server: EMQX

Topic: Node/red

QoS: Retain

User Properties: none

Response topic: none

Content Type: none

Expiry (secs): none

Name: Name

Tip: Leave topic, qos or retain blank if you want to set them via msg properties.

debug

all nodes all

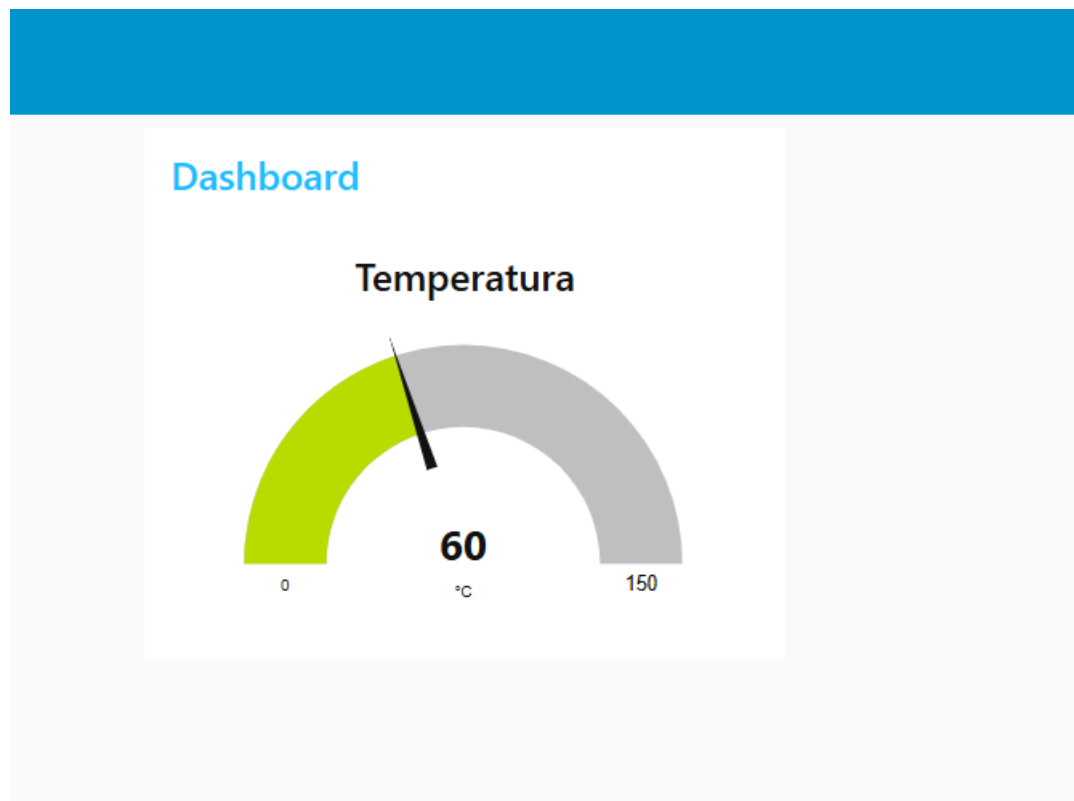
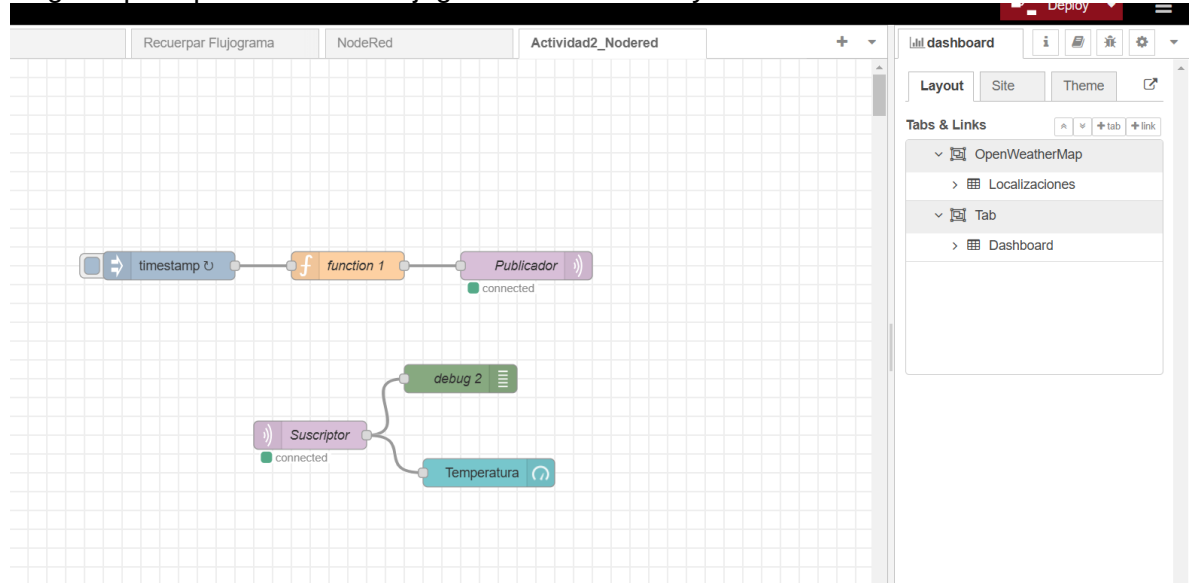
7/11/2024, 10:24:13 p. m. node: debug 1

Node/red : msg.payload : number

1731036252924

Actividad 2:

Pegar aquí el pantallazo del flujograma en node red y la visualización del dashboard



Actividad 3:

Pegar aquí el pantallazo de la Fig. 28 de la practica donde debe conectarse a una ciudad diferente a la de la practica y mostrar el dashboard para medir la temperatura de esa ciudad

NodeRed

Deploy

Edit openweathermap node

Delete

Cancel

Done

Properties

API Key

.....

Language

Spanish

Current weather for

Location

City, Country

City

Bucaramanga

Country

Colombia

Name

Name

Tip: you may want to manually search for the location first to check it can be found. Often two character country codes work best.

Note: The API Key may not immediately ready to use after creation. As FAQ said: Your API key will be activated automatically, between 10 minutes and 2 hours after your successful registration. Just wait a little bit.

Enabled

debug

all nodes

all

11/11/2024, 18:49:47 node: debug 1
Node/red : msg.payload : Object
{ temperature: 93 }
11/11/2024, 18:49:47 node: debug 3
msg.payload : Object
object
id: 741
weather: "Fog"
detail: "bruma"
icon: "50n"
tempk: 293.88
tempc: 20.7
temp_maxc: 20.7
temp_minc: 20.7
humidity: 100
pressure: 1013
maxtemp: 293.88
mintemp: 293.88
windspeed: 1.54
winddirection: 310
location: "Bucaramanga"
sunrise: 1731321733
sunset: 1731364264
clouds: 75
description: "The weather in Bucaramanga at coordinates: 7.1254..."

Recuperar Flujoograma

NodeRed

Actividad2_Nodered

Flow 1

marca tiempo

Tunja

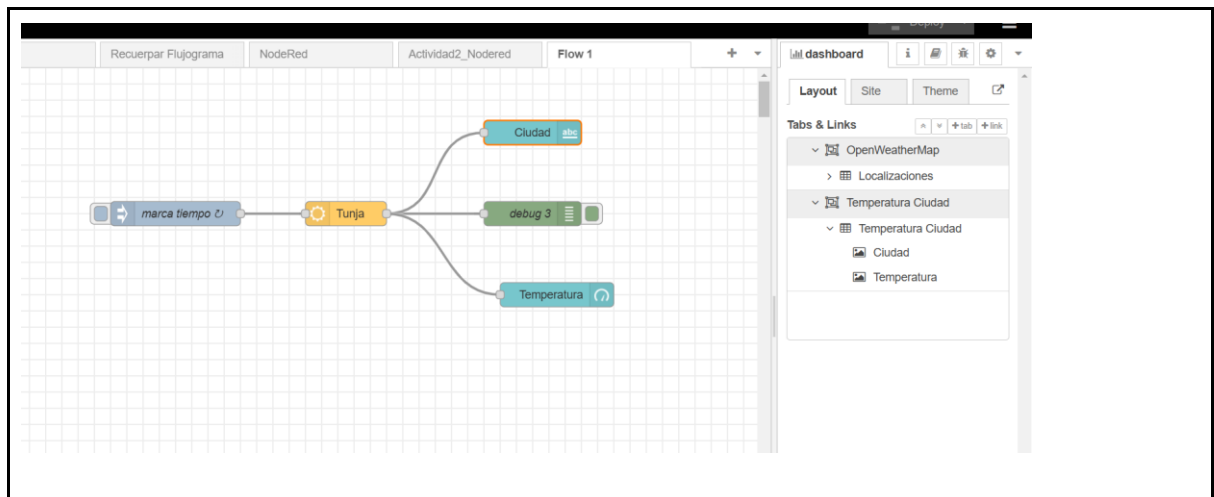
debug 3

debug

all nodes

all

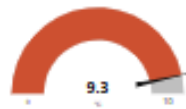
11/11/2024, 18:53:08 node: debug 1
Node/red : msg.payload : Object
{ temperature: 31 }
11/11/2024, 18:53:08 node: debug 3
msg.payload : Object
object
id: 804
weather: "Clouds"
detail: "nubes"
icon: "04n"
tempk: 282.53
tempc: 9.3
temp_maxc: 9.3
temp_minc: 9.3
humidity: 98
pressure: 1016
maxtemp: 282.53
mintemp: 282.53
windspeed: 1.36
winddirection: 143
location: "Tunja"
sunrise: 1731321670
sunset: 1731364445
clouds: 100



Temperatura Ciudad

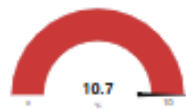
Ciudad
Tunja

Temperatura



Ciudad
Bogota

Temperatura



Ciudad
Bucaramanga

Temperatura



Ciudad
Santiago de Cali

Temperatura



Ciudad
Medellin

Temperatura



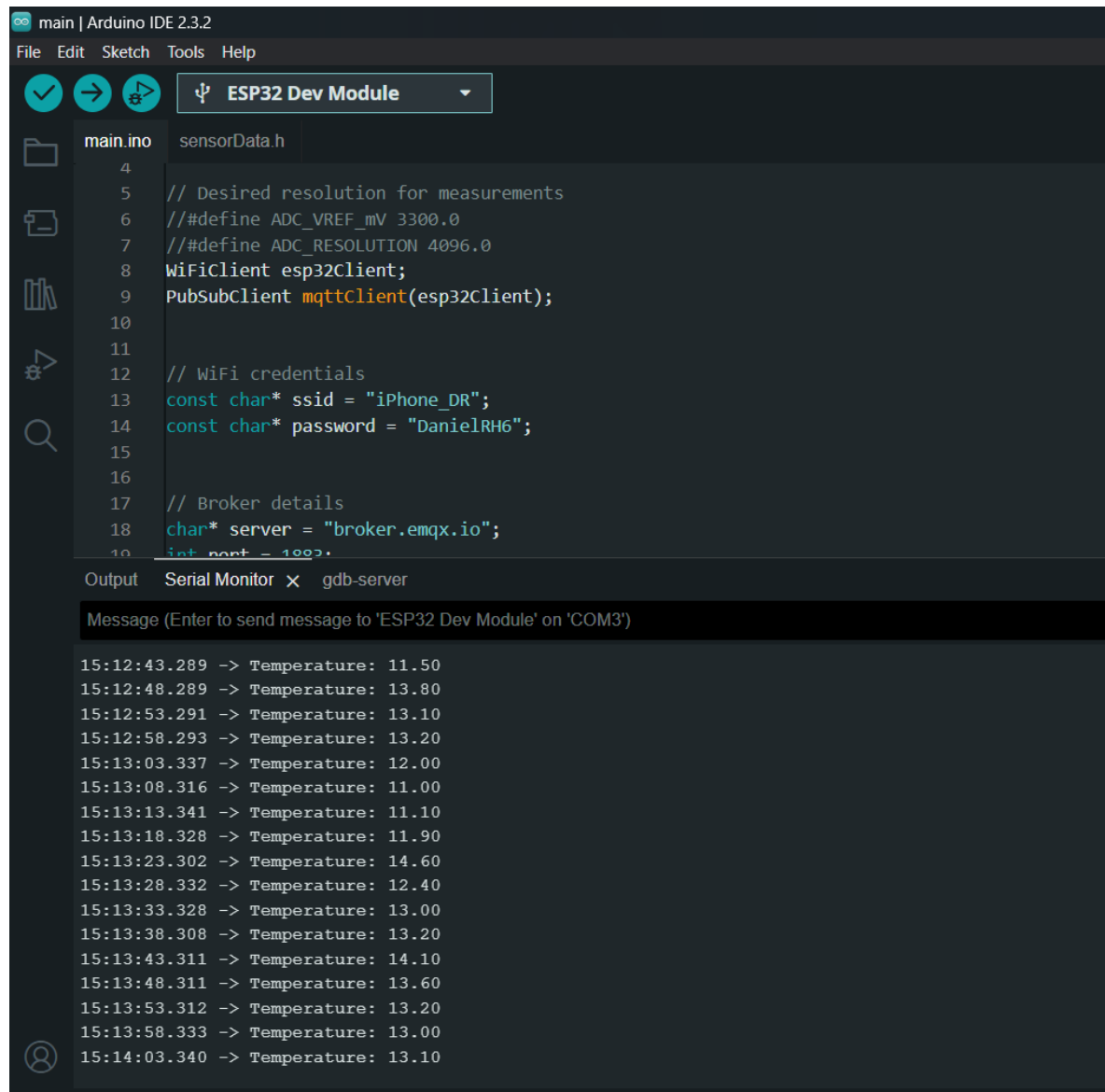
Ciudad
Cartagena

Temperatura



Actividad 4: RETO

Pegar aquí el flujograma propuesto en node red para la recepción de los datos y el dashboard para la visualización de las 7 ciudades señalando cuál es su ciudad



```
main | Arduino IDE 2.3.2
File Edit Sketch Tools Help
ESP32 Dev Module
main.ino sensorData.h
4
5 // Desired resolution for measurements
6 // #define ADC_VREF_mV 3300.0
7 // #define ADC_RESOLUTION 4096.0
8 WiFiClient esp32Client;
9 PubSubClient mqttClient(esp32Client);
10
11
12 // WiFi credentials
13 const char* ssid = "iPhone_DR";
14 const char* password = "DanielRH6";
15
16
17 // Broker details
18 char* server = "broker.emqx.io";
19 int port = 1883;
20
Output Serial Monitor x gdb-server
Message (Enter to send message to 'ESP32 Dev Module' on 'COM3')
15:12:43.289 -> Temperature: 11.50
15:12:48.289 -> Temperature: 13.80
15:12:53.291 -> Temperature: 13.10
15:12:58.293 -> Temperature: 13.20
15:13:03.337 -> Temperature: 12.00
15:13:08.316 -> Temperature: 11.00
15:13:13.341 -> Temperature: 11.10
15:13:18.328 -> Temperature: 11.90
15:13:23.302 -> Temperature: 14.60
15:13:28.332 -> Temperature: 12.40
15:13:33.328 -> Temperature: 13.00
15:13:38.308 -> Temperature: 13.20
15:13:43.311 -> Temperature: 14.10
15:13:48.311 -> Temperature: 13.60
15:13:53.312 -> Temperature: 13.20
15:13:58.333 -> Temperature: 13.00
15:14:03.340 -> Temperature: 13.10
```

Flow 2

Recuperar Flujos

NodeRed

Actividad2_Node

Flow 1

Flow 3

+

debug 11

Suscriptor ESP32
connected

Ciudad abc

Temperatura

Temperatura Ciudad

Temperatura Ciudad

Temperatura

0

13

30

°C

Ciudad:
Paris